V2: Decision-Driven Upgrade

From proxy classifier to operational targeting system

Built on frozen V1 baseline · CPS Census 1994-1995 · 199,523 records

Why V2 Was Needed

V1 proved a sound two-layer methodology: CatBoost classifier (PR-AUC 0.70) + 3-segment K-Prototypes. V2 addresses three structural limitations:

Proxy Target
V1 TREATS INCOME >50K AS THE ONLY FRAMING

3 Fixed Modes
V1 THRESHOLDS ARE METRIC-DRIVEN, NOT BUDGET-DRIVEN

47% Segment
V1 WORKING-AGE MACRO SEGMENT IS TOO BROAD

V2 adds a target hierarchy, budget-aware score bands, exact-lineage Layer 2 sub-segmentation, and optional extension validation — all without changing V1.

Target Hierarchy

TIER	TARGET	AVAILABLE?	STATUS
0	**Binary income >$50K**	**Yes**	V1 Baseline
1	Composite affluence / financial-activity indicator	Constructible	Designed
2	Response propensity / conversion / CLV	No (needs business data)	Future
3	E[value] = P(response) × E[revenue]	No	Ideal

Key insight

The biggest improvement opportunity is better targets, not better algorithms. V1's classifier is already strong (PR-AUC 0.70). Moving from Tier 0 to Tier 2 would be more impactful than any model-level tuning.

Score-Band Analysis

V2 replaces V1's 3 fixed thresholds with empirical score bands:

BAND	SCORE	N	% POP	POSITIVES	PRECISION
A: Very High	≥0.50	1,133	3.8%	876	77.3%
B: High	0.30–0.50	739	2.5%	298	40.3%
C: Marginal	0.15–0.30	1,209	4.0%	277	22.9%
D: Low	0.05–0.15	2,672	8.9%	230	8.6%
E: Very Low	<0.05	24,176	80.8%	176	0.7%

Band C is the decision zone. 81% of the population is confidently non-target (Band E). Bands A+B capture 63% of positives in 6.3% of population. Band C is where budget, capacity, and segment context determine the targeting decision.

Budget Knee

TOP K	TP	PRECISION	RECALL
500	463	92.6%	24.9%
1,000	798	79.8%	43.0%
1,500	**1,036**	**69.1%**	**55.8%**
2,000	1,214	60.7%	65.4%

5,000	1,639	32.8%	88.3%

Budget knee at ~1,500-2,000 contacts. Marginal TP per contact drops from 0.93 (first 500) to 0.10 (3,000-5,000).

Layer 2: Working-Age Sub-Segments

Conditionally Adopted — Run on exact V1 Segment 2 (91,443 rows). Lineage verified.

Established Earners

38% of WA · 20.7% >50K · 1.65x

- Mean age 51.5, married 74%
- Exec/managerial 19%, professional 16%

Service Workers

11% of WA · 2.4% >50K · 0.19x

- Mean age 33.6, female 56%
- Other service 17%, admin 15%

Younger Workers

51% of WA · 8.6% >50K · 0.69x

- Mean age 31.1, mixed occupations
- Bachelor's 18%, precision prod 11%

9x income-rate differential

Within the V1 working-age macro segment, income rate ranges from 2.4% (service workers) to 20.7% (established earners). The macro segment was hiding significant internal heterogeneity.

Full Architecture

```
Population (199,523)
  ├── Seniors (40,618, 20%)            → retirement
  ├── Youth (67,462, 34%)              → family-indirect
  └── Working-Age (91,443, 46%)
        ├── Established Earners (35K)   → premium targeting
```

```
        ├── Service Workers (10K)       → accessible products
        └── Younger Workers (46K)       → growth / mid-tier
```

Stability: 3/4 seeds produce 3 sub-segments. Service-worker group is most robust. Boundaries are directional.

Optional Extensions

Weighted Training

VARIANT	PR-AUC	VS BASELINE	DECISION
Baseline (no weight)	0.6932	–	Accepted
Normalized	0.6896	-0.0036	Not adopted
Sqrt	0.6887	-0.0045	Not adopted
Capped p99	0.6876	-0.0056	Not adopted

Classifier Benchmark

MODEL	VAL PR-AUC	TEST PR-AUC	DECISION
CatBoost	0.6932	0.7008	Retained
LightGBM	0.6839	0.6917	Not adopted
XGBoost	0.6829	–	Not adopted

CatBoost's native categorical handling gives it a structural advantage on this high-categorical dataset. No displacement.

Final V2 Architecture

Census Data → CatBoost Score → Score Band → Budget/Capacity Gate → Segment → Marketing Action

0.70	0.96	5
TEST PR-AUC	TEST ROC-AUC	SCORE BANDS

3+3

MACRO + MICRO SEGS

Governance Summary

CONCERN	STATUS
Weight excluded from features	Verified across all stages
Target excluded from clustering	Verified — profiling only
Survey-weighted evaluation	All metrics reported under both regimes
Numeric-coded categoricals	5 columns cast to string; never treated as continuous
Calibration	Assessed, not needed (Brier 0.031)
Fairness	Demographic features documented; governance note provided
Layer 2 lineage	Exact V1 labels recovered and verified
Layer 2 stability	3/4 seeds stable; caveat documented
Classifier diagnostics	PR-AUC by segment (rules-based assignment, caveat noted)

V2 Decision-Driven Upgrade — JPMC Take-Home Project